

Rafael Dutra

Applied AI Developer — Machine Learning & Automation

Recife, PE | +55 (81) 99965-4790 | rafael Dutra pro@gmail.com | linkedin.com/in/rafaelsantoshome

PROFESSIONAL SUMMARY

Applied AI Developer with hands-on experience in Machine Learning, Computer Vision, NLP, LLMs, and production AI pipelines. Currently working on AI solutions for education and security, including computer vision models, handwriting/drawing analysis, and integration with LLMs and NLP. C2 English certified by EFSET.

TECHNICAL SKILLS

AI & Machine Learning: LLMs (GPT-4o) · RAG · Autonomous Agents · NLP · Prompt Engineering · Computer Vision · InsightFace · Scikit-learn · TensorFlow · PyTorch · XGBoost · SHAP · LangChain
Automation & Infrastructure: FastAPI · Docker · Docker Compose · Redis · Microservices · PostgreSQL · MongoDB · REST APIs · Git/GitHub · Linux
Languages & Tools: Python · SQL · JavaScript · React · Pandas · NumPy · Jupyter

PROFESSIONAL EXPERIENCE

Applied AI Developer — FourDevs — Internship (Remote)

Apr/2026 — Present

- Developed AI solutions in Python, integrating Machine Learning models, inference pipelines, and APIs for production-ready applications.
- Built end-to-end pipelines, from model training and validation to deployment through FastAPI in a Docker-based production environment.

AI Developer — Porto Digital / Globo Technological Residency — Recife, PE

Feb/2024 — Present

- Horus AI: developed 3 ML models — video 97.6%, audio 90.9%, lipsync 100% — for automatic anomaly detection in live broadcasts, replacing manual monitoring with real-time detection.
- Built video collection and processing pipelines for MP4, WMV, HLS/SRT streams, integrating Angular frontend, FastAPI backend, and ML models in a Docker environment.

FEATURED PROJECTS

Horus AI — Streaming Anomaly Detection • Python · FastAPI · TensorFlow · OpenCV · Angular · WebSocket

2025

- Designed a hybrid OpenCV + ML strategy with temporal voting, automatic evidence clip generation, WebSocket alerts, KPI dashboard, and PDF report export.

OpenFinance Intel — Financial Intelligence Platform • Python · FastAPI · Redis · MongoDB · Docker · NLP

2025

- Built a continuous ingestion pipeline using RSS and Reddit, with real-time processing, normalization, and deduplication before persistence in MongoDB.
- Applied NLP and prompt engineering for analysis, classification, and summarization of financial market text events.

EDUCATION

Systems Analysis and Development — FICR — Faculdade Imaculada Conceição do Recife

Feb/2024 — Jul/2026

Relevant coursework: Data Structures, Databases, Artificial Intelligence, Data Science, Technological Residency.

LANGUAGES & CERTIFICATIONS

Languages: Portuguese Native | English C2 Proficient — EFSET Certified | Basic Spanish

Certifications: EF SET C2 Proficient | ONE Tech Foundation G8 — Data Science + AI (Oracle/Alura) | SQL Advanced (Kaggle) | Python (Kaggle)

In progress: Microsoft AI-900 — Azure AI Fundamentals